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grupa 241

VARIABILE ALEATOARE DISCRETE
-TEMĂ-

1 Fie X și Y două v.a. discrete independente.

1) Pornind de la X și Y, construiți următoarele v.a:

a) $X: \begin{pmatrix} 2 & 3 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$ $Y: \begin{pmatrix} -3 & -2 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

- $3X = 3 \cdot X: \begin{pmatrix} 2 \cdot 3 & 3 \cdot 3 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix} = \begin{pmatrix} 6 & 9 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$

- $X^{-1}: \begin{pmatrix} 2^{-1} & 3^{-1} \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$

- $\underbrace{\cos\left(\frac{\pi}{2} \cdot X\right)}_{\text{funcții}}: \begin{pmatrix} \cos\left(\frac{\pi}{2} \cdot 2\right) & \cos\left(\frac{\pi}{2} \cdot 3\right) \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$

- $Y^2: \begin{pmatrix} (-3)^2 & (-2)^2 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

- $Y + 3: \begin{pmatrix} -3+3 & -2+3 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$.

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b) $X: \begin{pmatrix} 0 & 9 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ $Y: \begin{pmatrix} -3 & 1 \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix}$

- $X-1: \begin{pmatrix} 0-1 & 9-1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 8 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$
- $X^{-2}: \begin{pmatrix} 0^{-2} & 9^{-2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{81} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$
- $\sin\left(\frac{\pi}{4} \cdot X\right): \begin{pmatrix} \sin\left(\frac{\pi}{4}\cdot 0\right) & \sin\left(\frac{\pi}{4}\cdot 9\right) \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \sin 0 & \sin\frac{\pi}{4} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} =$

$= \begin{pmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

- $Y \cdot 5: \begin{pmatrix} -3\cdot 5 & 1\cdot 5 \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix} = \begin{pmatrix} -15 & 5 \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix}$
- e^{Y} (funcție) $\Leftrightarrow \begin{pmatrix} e^{-3} & e^{1} \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix} = \begin{pmatrix} \frac{1}{e^3} & e \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix}$, $e \simeq 2{,}71$

c) $X: \begin{pmatrix} 5 & 8 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$ $Y: \begin{pmatrix} -1 & 1 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}$

- $2X: \begin{pmatrix} 5\cdot 2 & 8\cdot 2 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} = \begin{pmatrix} 10 & 16 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

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$\cdot X^{-3}: \begin{pmatrix} 5^{-3} & 8^{-3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{125} & \frac{1}{512} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$\cdot tg(\pi X): \begin{pmatrix} tg(5\pi) & tg(8\pi) \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{sin(5\pi)}{cos(5\pi)} & \frac{sin(8\pi)}{cos(8\pi)} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} =$

$= \begin{pmatrix} 0 & 0 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}.$

$\cdot Y-2: \begin{pmatrix} -1-2 & 1-2 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix} = \begin{pmatrix} -3 & -1 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}$

$\cdot |Y|: \begin{pmatrix} |-1| & |1| \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}.$

d) $X: \begin{pmatrix} -3 & 6 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix} \quad Y: \begin{pmatrix} e & e^3 \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$

$\cdot 2-X: \begin{pmatrix} (-3\cdot(-1))+2 & (6\cdot(-1))+2 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix} = \begin{pmatrix} 5 & -4 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix}$

$\cdot X^3: \begin{pmatrix} -3^3 & 6^3 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix} = \begin{pmatrix} -27 & 216 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix}.$

$\cdot \cos\left(\frac{\pi}{6}\cdot X\right): \begin{pmatrix} \cos\left(\frac{\pi}{62}\cdot(-3)\right) & \cos\left(\frac{\pi}{6}\cdot 6\right) \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ \frac{1}{8} & \frac{7}{8} \end{pmatrix}$

$\cdot Y^{-1}: \begin{pmatrix} e^{-1} & e^{-3} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{e} & \frac{1}{e^3} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}.$

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• $\ln Y : \begin{pmatrix} \ln e & \ln e^3 \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$.

2) Folosind v. a X și Y de la 1) determinați:

a) • $2X + 3Y$;

$2X : \begin{pmatrix} 4 & 6 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$; $3Y : \begin{pmatrix} -9 & -6 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

$\Rightarrow 2X+3Y : \begin{pmatrix} 4-9 & 4-6 & 6-9 & 6-6 \\ \frac{1}{5}\cdot\frac{4}{5} & \frac{1}{5}\cdot\frac{1}{5} & \frac{4}{5}\cdot\frac{4}{5} & \frac{4}{5}\cdot\frac{1}{5} \end{pmatrix} =$

$= \begin{pmatrix} -5 & -2 & -3 & 0 \\ \frac{4}{25} & \frac{1}{25} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}$

• $3X - Y$;

$3X : \begin{pmatrix} 6 & 9 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$ $Y : \begin{pmatrix} -3 & -2 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

$3X - Y : \begin{pmatrix} 6+3 & 6+2 & 9+3 & 9+2 \\ \frac{1}{5}\cdot\frac{4}{5} & \frac{1}{5}\cdot\frac{1}{5} & \frac{4}{5}\cdot\frac{4}{5} & \frac{4}{5}\cdot\frac{1}{5} \end{pmatrix} =$

$= \begin{pmatrix} 9 & 8 & 12 & 11 \\ \frac{4}{25} & \frac{1}{25} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}$.

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• $X^2 \cdot Y^3$;

$X^2: \begin{pmatrix} 4 & 9 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$ $\quad Y^3: \begin{pmatrix} -27 & -8 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

$X^2 Y^3: \begin{pmatrix} 4\cdot(-27) & 4\cdot(-8) & 9\cdot(-27) & 9\cdot(-8) \\ \frac{1}{5}\cdot\frac{4}{5} & \frac{1}{5}\cdot\frac{1}{5} & \frac{4}{5}\cdot\frac{4}{5} & \frac{4}{5}\cdot\frac{1}{5} \end{pmatrix} =$

$= \begin{pmatrix} -108 & -32 & -243 & -72 \\ \frac{4}{25} & \frac{1}{5} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}.$

b) • $X - Y: \begin{pmatrix} 0+3 & 0-1 & 9+3 & 9-1 \\ \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} & \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} \end{pmatrix} =$

$= \begin{pmatrix} 3 & -1 & 12 & 8 \\ \frac{1}{14} & \frac{6}{14} & \frac{1}{14} & \frac{6}{14} \end{pmatrix}.$

• $\cos(\overline{4} \times Y)$;

$X \cdot Y: \begin{pmatrix} 0\cdot(-3) & 0\cdot 1 & 9\cdot(-3) & 9\cdot 1 \\ \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} & \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} \end{pmatrix} = \begin{pmatrix} 0 & 0 & -27 & 9 \\ \frac{1}{14} & \underbrace{\frac{6}{14}}_{(+)} & \frac{1}{14} & \frac{6}{14} \end{pmatrix}$

$= \begin{pmatrix} 0 & -27 & 9 \\ \frac{4}{14} & \frac{1}{14} & \frac{6}{14} \end{pmatrix}.$

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$\cos(\pi XY): \begin{pmatrix} \cos(0) & \cos(-2\pi) & \cos(9\pi) \\ \frac{7}{14} & \frac{1}{14} & \frac{6}{14} \end{pmatrix} =$

$= \begin{pmatrix} 1 & -1 & -1 \\ \frac{7}{14} & \frac{1}{14} & \frac{6}{14} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ \frac{7}{14} & \frac{7}{14} \end{pmatrix}$

$(+)$

- $X^2 + 3Y$;

$X^2: \begin{pmatrix} 0 & 81 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \qquad 3Y: \begin{pmatrix} -9 & 3 \\ \frac{1}{7} & \frac{6}{7} \end{pmatrix}$

$X^2 + 3Y: \begin{pmatrix} 0-9 & 0+3 & 81-9 & 81+3 \\ \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} & \frac{1}{2}\cdot\frac{1}{7} & \frac{1}{2}\cdot\frac{6}{7} \end{pmatrix} =$

$= \begin{pmatrix} -9 & 3 & 72 & 84 \\ \frac{1}{14} & \frac{6}{14} & \frac{1}{14} & \frac{6}{14} \end{pmatrix}$

c) $X + Y: \begin{pmatrix} 5-1 & 5+1 & 8-1 & 8+1 \\ \frac{1}{3}\cdot\frac{1}{6} & \frac{1}{3}\cdot\frac{5}{6} & \frac{2}{3}\cdot\frac{1}{6} & \frac{2}{3}\cdot\frac{5}{6} \end{pmatrix} =$

$= \begin{pmatrix} 4 & 6 & 7 & 9 \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix}$

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· $\sin\left(\frac{\pi}{2} \times y\right)$;

$$X \cdot Y : \begin{pmatrix} 5\cdot(-1) & 5\cdot 1 & 8\cdot(-1) & 8\cdot 1 \\ \frac{1}{3}\cdot\frac{1}{6} & \frac{1}{3}\cdot\frac{5}{6} & \frac{2}{3}\cdot\frac{1}{6} & \frac{2}{3}\cdot\frac{5}{6} \end{pmatrix} = \begin{pmatrix} -5 & 5 & -8 & 8 \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix}$$

$$\sin\left(\frac{\pi}{2} \times y\right) : \begin{pmatrix} \sin\left(\frac{-5\pi}{2}\right) & \sin\left(\frac{5\pi}{2}\right) & \sin\left(\frac{-8\pi}{2}\right) & \sin\left(\frac{8\pi}{2}\right) \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix} =$$

$$= \begin{pmatrix} -1 & 1 & 0 & 0 \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix}$$

· $\frac{1}{X} + \frac{1}{Y} \Leftrightarrow X^{-1} + Y^{-1}$

$$X^{-1} : \begin{pmatrix} \frac{1}{5} & \frac{1}{8} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} \qquad Y^{-1} : \begin{pmatrix} -1 & 1 \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}$$

$$X^{-1} + Y^{-1} : \begin{pmatrix} \frac{1}{5} \overset{5}{-} 1 & \frac{1}{5} \overset{5}{+} 1 & \frac{1}{8} \overset{8}{-} 1 & \frac{1}{8} \overset{8}{+} 1 \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{4}{5} & \frac{6}{5} & -\frac{7}{8} & \frac{9}{8} \\ \frac{1}{18} & \frac{5}{18} & \frac{2}{18} & \frac{10}{18} \end{pmatrix}$$

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4) Folosind rezultatele de la 1), 2) calculați:

a) $\mathbb{P}(2X+3Y>1)$;

$$2X+3Y:\begin{pmatrix} -5 & -2 & -3 & 0 \\ \frac{4}{25} & \frac{1}{25} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}$$

În acest caz, $\mathbb{P}(2X+3Y>1)=0$

- $\mathbb{P}(2X+3Y>1 \mid X>0) = \mathbb{P}(2X+3Y>1)=0$.

evenimente independente

- $\mathbb{P}(2X+3Y<3 \mid Y<-2) = \frac{\mathbb{P}((2X+3Y<3)\cap(Y<-2))}{\mathbb{P}(Y<-2)} =$ ($\cap$ = și)

$= \frac{\mathbb{P}(2X+3Y<3)\cdot 1}{\mathbb{P}(Y<-2)} = \frac{\mathbb{P}(2X+3Y<3)}{\frac{4}{5}}$

$= \frac{\mathbb{P}(2X+3Y<3)\cdot\mathbb{P}(Y<-2)}{\mathbb{P}(Y<-2)} = \frac{1\cdot\frac{4}{5}}{\frac{4}{5}} = 1.$

- $\mathbb{P}(X^2Y^3>3)$;

$$X^2Y^3:\begin{pmatrix} -108 & -32 & -243 & -72 \\ \frac{4}{25} & \frac{1}{5} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}.$$

Nicio $*$ pentru care $X^2Y^3>3$ să fie adevărată

$\Rightarrow \mathbb{P}(X^2Y^3>3)=0.$

- $\mathbb{P}(X^2Y^3\leq 3)=\frac{4}{25}+\frac{1}{5}+\frac{16}{25}+\frac{4}{25}=1.$

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. $\mathbb{P}(2X + 3Y < 3X - Y)$;

$\mathbb{P}(2X - 3X + 3Y + Y < 0) \Leftrightarrow \mathbb{P}(-X + 4Y < 0)$

Calculăm $-X + 4Y$, știind că X, Y sunt v.a independente.

$-X : \begin{pmatrix} -2 & -3 \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$, $4Y : \begin{pmatrix} -12 & -8 \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix}$

$-X + 4Y : \begin{pmatrix} -14 & -10 & -15 & -11 \\ \frac{4}{25} & \frac{1}{25} & \frac{16}{25} & \frac{4}{25} \end{pmatrix}$

În acest caz, $\mathbb{P}(-X + 4Y < 0) = \frac{4}{25} + \frac{1}{25} + \frac{16}{25} + \frac{4}{25} =$

$(=1)$. ??

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3) Determinați parametrii reali p și q, știind că:

a) $X: \begin{pmatrix} 1 & 2 \\ p & q \end{pmatrix}$ $\quad Y: \begin{pmatrix} 3 & 9 \\ 0,1 & \frac{p^2+0,02}{2} \end{pmatrix}$

sunt bine definite

îl scoatem pe p din Y

C.E. $\frac{p^2+0,02}{2} \geq 0 \Leftrightarrow p^2+0,02 \geq 0 \Leftrightarrow p^2 \geq -0,02$

$\Rightarrow \nexists$ soluții pentru p. !

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